M&AE 5070 - Flight Dynamics

Case Study 7 - SOLUTIONS

Lateral Dynamics

1 Introduction

In this case study, we study the responses of a civil transport aircraft to changes in lateral controls, $\Delta\delta_\ell$ and $\Delta\delta_n$, and to initial side-slip angles and bank angle. The initial equilibrium state is a straight, level flight at low velocity and altitude ($V = 125$ m/s and $h \approx 0$ m). The analysis of the temporal response of the aircraft will show the natural lateral modes of the aircraft (roll, dutch roll and spiral). Strategies to stabilize the dutch roll will be explored.

2 Available data

The aircraft data is representative of a long-haul, 4-engine, 200-passengers aircraft. The main reference parameters are provided below:

- $\ell = 6.5$ m
- $d = 20$ m
- $m = 1.2 \times 10^5$ kg
- $S = 240$ m^2
- $A = 5.88 \times 10^6$ kg.m^2
- $C = 11.1 \times 10^6$ kg.m^2

The lateral coefficients are:

Coefficients	C_Y	C_ℓ	C_n
β	$C_{Y_\beta} = -0.65$	$C_{\ell_\beta} = -0.92$	$C_{n_\beta} = 0.8175$
p	$C_{Y_p} = 0$	$C_{\ell_p} = -18.6$	$C_{n_p} = -1.37$
r	$C_{Y_r} = 0$	$C_{\ell_r} = 5.89$	$C_{n_r} = -7.18$
δ_ℓ	$C_{Y_{\delta_\ell}} = 0$	$C_{\ell_{\delta_\ell}} = -0.56$	$C_{n_{\delta_\ell}} = -0.02$
δ_n	$C_{Y_{\delta_n}} = 0.19$	$C_{\ell_{\delta_n}} = 0.13$	$C_{n_{\delta_n}} = -0.56$

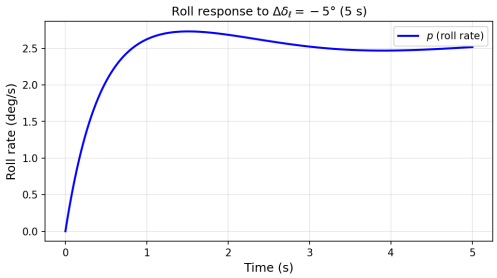
1. Write a script that integrates the lateral motion equations above, make sure you can easily input initial conditions for side-slip, ailerons and rudder. Use it to do the following:

   (a) Plot the roll angular velocity of the aircraft as a function of time for a delta in roll control, $\Delta\delta_\ell = -5^o$(a 5s integration time is appropriate). Estimate p_∞ and characteristic time scale fo the roll mode from your graph and compare them to theory.

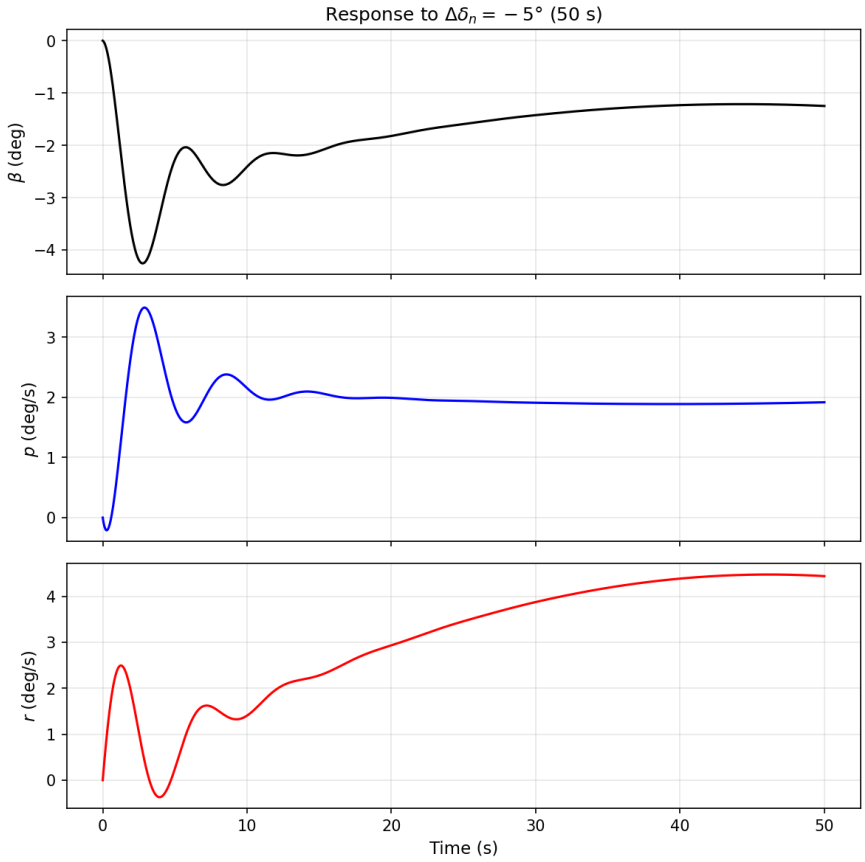
   (b) Plot side slip angle, roll angular velocity and yaw angular velocity of the aircraft as a function of time for a delta in yaw control, $\Delta\delta_n = -5^o$(a 50s integration time is appropriate). Comment on what you see, and compare the period of the oscillations to the proper theoretical quantity.

   Solutions.

   - **Roll mode.** This mode can be observed from the p temporal response to a $\Delta\delta_\ell = -5^o$, as shown below. We can get a rough estimate of the roll mode time constant from it, around 0.3 s. The analytical value gives a value of 0.4s. The steady state value p_∞ is around 2.6 deg/s from graph, compared to a theoretical value of 2.9 deg/s. Note that the roll mode is not perfectly decoupled from the rest of the equations, as evidenced by the slight super-imposed oscillatory behavior visible on the plot.



- **Dutch roll mode.** The Dutch roll mode can be observed when applying a $\Delta\delta_n = -5°$control. The period can be estimated both from the graph and theory at around 6s.



A couple things to note on the graphs:

- The roll mode is excited by the oscillations between side slip angle and yaw angular velocity.
- No steady state is observed : at longer times, the oscillations from the dutch roll have been damped, but yaw velocity and side slip do not go back to zero. Instead, they very slowly reach a non-zero asymptote, with the bank angle continuously increasing. This can be interpreted as the aircraft getting into a turn, but without stabilizing it. In order to stabilize it and get a steady turn, the pilot would need to act again on the stick (and pedals potentially) to recover the conditions corresponding to a steady turn.

2. Start a Dutch roll by applying an initial side-slip angle $\beta_i = 4°$. We want to analyze how the "route stability" coefficient C_{n_β}, the dihedral effect coefficient C_{ℓ_β}, and the altitude affect the mode.

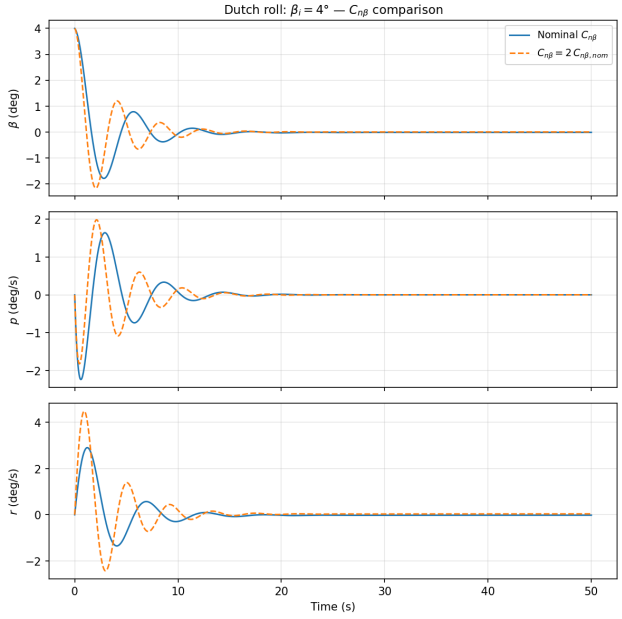
   (a) **Route stability**: Compare the temporal response obtained using the nominal C_{n_β} to a value twice as large.

(b) **Dihedral effect**: Compare the temporal response obtained using the nominal C_{ℓ_β} to a value twice as large.

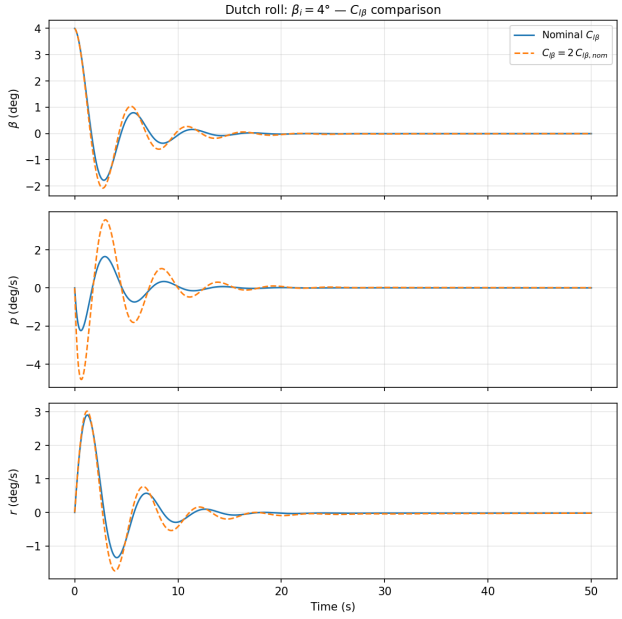
(c) **Altitude and flight conditions**: Compare the temporal response obtained using the nominal case ($V = 125m/s$ and $h = 0$) to a typical cruise condition ($V = 240m/s$ and $h = 12,000$)

Solutions.

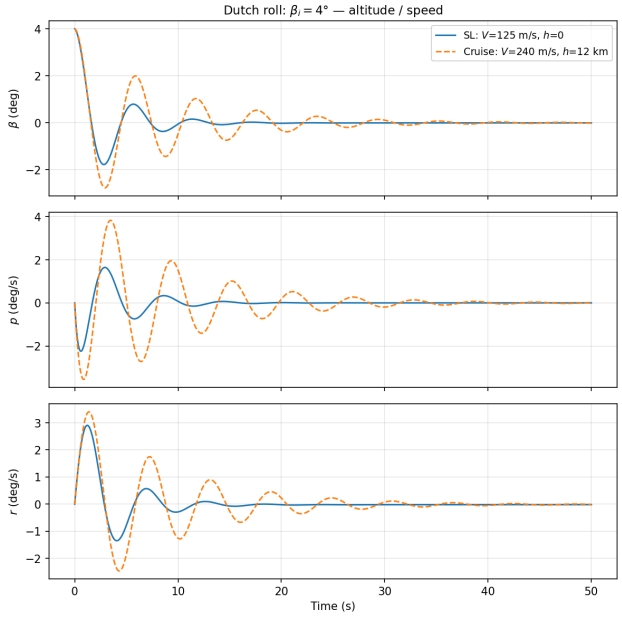
- Doubling C_{n_β} changes the frequency and the damping ratio of the dutch roll. Since the frequency is proportional to $\sqrt{C_{n_\beta}}$, the period should be roughly divided by $\sqrt{2}$. Same thing for ζ, which translate into more visible oscillations. However, the damping itself remains roughly constant: the envelope of the curves does not change significantly when changing C_{n_β}.



- Doubling C_{ℓ_β} (dihedral effect) does not change the dutch roll itself, however, it significantly impact p (this illustrates the fact that p does not feedback significantly in the evolution of β and r).



- The conditions provided roughly satisfy $\frac{1}{2}\rho V^2 =$ constant: this change of flight condition from low V, low altitude to high V, high altitude, will therefore not change the frequency of the dutch roll. The damping is independent of the velocity, but depends on $\sqrt{\rho}$: we should expect to see it decrease by a factor of roughly $\sqrt{3}$, since the density at 12km is roughly 1/3 of the sea level density.



3. You are in charge of tuning the Dutch roll to ensure that its damping ratio $\zeta = 0.7$. Design and implement an automatic control law to achieve this objective. You can evaluate the gain of the feedback loop analytically, or "experimentally", trying out various values in MATLAB. Test your control on an initial side-slip of 5°.

   Solutions.

   A simple control law of the form $\delta_n = 1.8r$ allows for the damping ratio to be increased from around 0.3 to 0.7. Note however that it has a clear effect on the spiral mode, going from a time constant of around -700 s to 17s. If the spiral mode is too fast, it may become annoying from the pilot point of view, so should be kept in mind.

4. Start a spiral by applying an initial bank angle $\phi_i = 30°$, and analyze how the "route stability" coefficient C_{n_β}, the dihedral effect coefficient C_{ℓ_β} affect the mode.

   (a) **Route stability**: Compare the nominal C_{n_β} to a value twice as large. What is the value of C_{n_β} that destabilizes the spiral mode? Check with the simulations.

   (b) **Dihedral effect**: Compare the nominal C_{ℓ_β} to a value twice as large.

   Solutions.

   Doubling C_{n_β} destabilizes the spiral mode, since the criterion for stability is

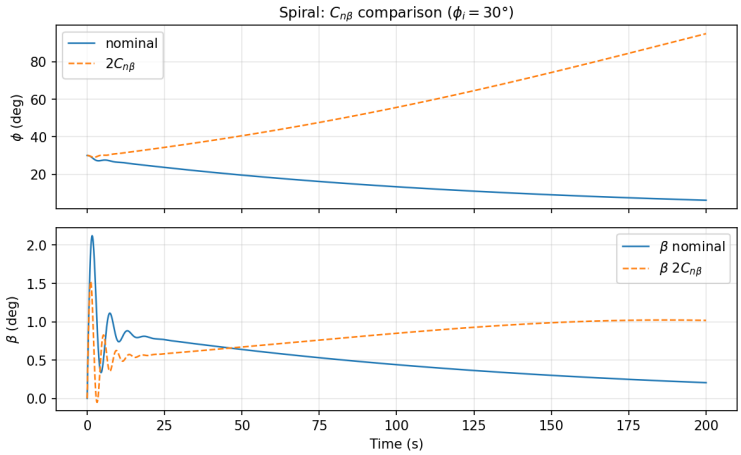
$$C_{\ell_\beta} C_{n_r} > C_{\ell_r} C_{n_\beta} \tag{1}$$

   If the term $C_{\ell_r} C_{n_\beta}$ increases, the criterion cannot be satisfied anymore. More specifically, the spiral mode becomes unstable when:

$$C_{\ell_\beta} C_{n_r} = C_{\ell_r} C_{n_\beta} \tag{2}$$

$$(-0.92)(-7.18) = 5.89 \times n \times 0.8175 \tag{3}$$

where n is the factor by which C_{n_β} must be multiplied to reach the neutral point. We find $n = 1.37$: for the mode to become unstable, C_{n_β} must be increased by 37%.



With similar arguments, we can easily show that increasing C_{ℓ_β} increases the stability of the spiral mode.

